

Wicaksono Leksono Muhamad

AI / Machine Learning Engineer & Researcher

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EDUCATION

Universitas Gadjah Mada

Master of Artificial Intelligence

Yogyakarta, Indonesia

Feb. 2024 – Oct. 2025

Institut Teknologi Sepuluh Nopember

B.Sc. Computational and Applied Mathematics

Surabaya, Indonesia

Aug. 2018 – Nov. 2022

EXPERIENCE

Research Assistant, Software Engineering Lab

Universitas Gadjah Mada

Mar. 2025 – Oct. 2025

Yogyakarta, Indonesia

- Built LLM-based tooling for requirements engineering; added pragmatic and semantic testing on user story quality specifications, extending prior automated QUS analysis.
- Led ML workstream for Laptop Merah Putih 2025 research grant; delivered LLM-based PHQ screening with language model personalization and facial affect detection using Flask, evaluated on 302 samples.

Researcher, Natural Language Processing

Mantera Studio Research Lab

Feb. 2022 – Present

Jakarta, Indonesia

- Competing in BEA 2026 and SemEval 2026 shared tasks; achieved 93% accuracy on SemEval Task 11 benchmark for syllogistic reasoning evaluation (competition ongoing).
- Developed multilingual translation and parsing pipelines using Google Gemini API for cross-lingual vocabulary difficulty prediction across Spanish, Bengali, Russian, German, and other languages.

Web Developer

Mastej Studio

2019 – 2022

Malang, Indonesia

- Built *What's On Malang* (React/Vite + custom CMS); grew to 400+ daily active users and 100 new leads per day by digitalizing tourism content for the Malang region.
- Delivered Begawan Apartment marketing site with 360-degree virtual tour; generated 10 daily inquiries from outside the Malang region, expanding regional marketing reach.

Head of Student Resource Development

Institut Teknologi Sepuluh Nopember

Feb. 2020 – Nov. 2021

Surabaya, Indonesia

- Led 12 programs across the Student Association; achieved 97% KPI attainment against a 75% target.
- Delivered campus-scale events (MABACUP, UKM EXPO, Managerial School for Event Development) supporting 500+ participants.

PUBLICATIONS & RESEARCH

Aligning Implied Statements for Implicit Hate Speech Generalizability

First Author – Under Peer Review: ACL Rolling Review (OpenReview)

Jan. 2026

- Trained triplet objective with context-bounded semi-hard negatives to separate near-surface pairs in hate speech embeddings.
- Improved out-of-domain F1-score by 4–8% across 3 hate speech benchmarks over contrastive learning baselines.

LLM-Based Conversational PHQ-Aligned Screening (CPAS)

Person in Charge (ML) – Under Review: IEEE Access; ACM TiiS

Mar. 2025 – Dec. 2025

- Designed LLM-driven screening protocol aligned with PHQ-9 combining open-ended dialogue, questionnaire, and facial affect analysis.
- Validated against PHQ reference scores; achieved 0.6–0.8 Spearman's ρ correlation and 0.88 agreement on 151 samples.

Automated Detection of Semantic and Pragmatic Quality Issues in User Stories

Research Assistant – Presented: ICODSE 2025

Mar. 2025 – Jul. 2025

- Extended AQUASA framework with LLM-assisted checks for user story quality; introduced pragmatic and semantic testing capabilities absent in prior work.

On-Edge Device Optimization for Audio Classification

2025

First Author – Published: IEEE Xplore

- Deployed MFCC-based audio classifier on Arduino Nano 33 BLE; achieved 95.2% accuracy with 3.00 ms latency and 3.80 KB memory footprint.

Knowledge Distillation for GCN-Based Neural Networks

2022

Bachelor Thesis – Traffic Volume Prediction

- Achieved 50% model compression using 3 KD methods (feature-based via adapter, soft-loss, weight transfer) on ASTGCN; improved smaller model accuracy from 30% to 60% against 80% teacher baseline.

Multimodal Image–Table Pipeline

Jun. 2025

Multimodal Document Pipeline – Paper in Preparation

- Built document splitting and alignment pipeline for images, tables, and text using graph-based heuristics for structured extraction.

SOFTWARE PROJECTS

Mental Health Screening Platform | *Flask, LangChain, PyTorch, RPC* | mentalhealth.laptopmerahputih.id Nov. 2025

- Shipped PHQ + CPAS screening workflows; served 40+ concurrent user sessions with stable uptime.
- Integrated facial affect capture; reduced p95 latency and cost by 90% through local response batching and request queuing.

IndoClimate Chatbot | *React, LangChain, RAG* | chat.indoclimate.id

Aug. 2025

- Developed WhatsApp webhook + API with RAG over Indonesian climate change legislation; built LLM summarization pipeline for long legal documents to improve similarity matching.

TASYA Academic Chatbot | *Flask, LangChain* | tasya.dtmi.ugm.ac.id

Feb. 2025

- Built WhatsApp webhook + API for academic support RAG; developed LLM + heuristic document parsing pipeline for multimodal university documents.

Security Operations Centre Dashboard | *Vite.js, Redux Toolkit, HeadlessUI, Go*

Aug. 2024

- Refactored frontend into domain-driven design; reduced hardcoded and redundant code by 70%. Integrated server telemetry ingestion and LLM-based summarization.

TECHNICAL SKILLS

Programming Languages: Python, TypeScript/JavaScript, Go, Java, C/C++, Rust, SQL (PostgreSQL)

ML / AI Frameworks: PyTorch, TensorFlow, HuggingFace Transformers/Trainer, scikit-learn, LangChain

Web Frameworks: React, Next.js, Flask, Node.js, Vite.js

Data & Tools: Pandas, NumPy, Matplotlib, Docker, Git, Google Cloud Platform, Linux, L^AT_EX

Languages: Bahasa Indonesia (Native), English (Professional – TOEFL ITP 587), German (B1), Javanese (Conversational)